

Sandeep Kambhampati



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Education

Harvard University (Boston, MA)

Expected December 2026

Ph.D. candidate, Bioinformatics and Integrative Genomics

Johns Hopkins University (JHU) (Baltimore, MD)

May 2021

B.S., Biomedical Engineering (BME) | Genomics & Systems Biology Track | GPA: 3.93

Technical Skills

Proficient: Python, PyTorch, Lightning, Pyro, R, Unix, Git

Basic: JAX/NumPyro, Julia, Java

Experienced in Self-Supervised Learning, Generative Modelling, Computer Vision, Transformers, Pre-training, Bayesian Inference, Generalized Linear Models

Research Experience

Microsoft Research New England – *Research Intern*

Summer 2025

Advisor: Dr. Alex Lu (*Senior Researcher*)

- Designed *Vermeer*, a channel-adaptive autoregressive transformer that generates microscopy images of protein localization and generalizes to previously unseen proteins
- Applying *Vermeer* as a *virtual staining* model to investigate how pathogenic mutations alter protein localization and overcome limitations in experimental data collection

Broad Institute of MIT & Harvard – *Graduate Research Student*

2021-present

Advisors: Drs. Fei Chen (*Institute Member*), Mehrtash Babadi (*Director of Computational Methods*)

- Built *TissueMosaic*, a self-supervised representation learning framework for spatial transcriptomics (ST) data, by adapting DINO with domain-specific augmentations.
- Formulated a latent neural PDE framework to model how interactions between local spatial processes give rise to developmental patterns in time-series ST data
- Developing a Point Cloud Transformer foundation model to learn rules of tissue organization

Johns Hopkins University – *Undergraduate Researcher*

2017-2021

- Constructed a statistical physics model of DNA methylation from long read ONT sequencing data and applied this model to study genetic heritability of DNA methylation patterns
- Analyzed RNA-seq data to identify key transcriptional regulators of multi-organ development
- Engineered mouse knockout and *in vitro* overexpression models to identify gene targets that enhance the therapeutic potential of stem-cell derived cardiomyocytes

Publications (selected)

1. **Kambhampati, S.**, ... Chen, F.#, Lu, A.# Vermeer: Autoregressive generative modeling of microscopy predicts protein localization. *Under Review* (2026) [Github](#) [bioRxiv](#)
2. Bhate, S.*, **Kambhampati, S.***, Babadi, M., Chen, F.#, Uhler, C.# DALI Learns Rules Generating Spatiotemporal Transcriptomics. *ICLR GEN2 Workshop*. (2026) [Github](#) [Paper](#)
3. **Kambhampati, S.**, D'Alessio, L., Grab, F., Fleming, S., Chen, F.#, Babadi, M.# TissueMosaic enables cross-sample differential analysis of spatial transcriptomics datasets through self-supervised representation learning. *Cell Systems* (2025) [Github](#) [Paper](#)

4. **Kambhampati, S.***, Murphy, S.*, *et al.* Cross-organ transcriptomic comparison reveals universal factors during maturation. ***Journal of Computational Biology*** (2022)
5. Davidovich, A., ... **Kambhampati, S.**, *et al.* Non-Mendelian inheritance of DNA methylation patterns in mice. ***Nature Genetics*** (2026)
6. Russell, A.*, Weir, J.*, Nadaf, N.* ... **Kambhampati, S.**, *et al.* Slide-tags enables single-nucleus barcoding for multimodal spatial genomics. ***Nature*** (2023)
7. Abanti, J., **Kambhampati, S.**, Feinberg, A., Goutsias, J. Estimating DNA methylation potential energy landscapes from nanopore sequencing data. ***Scientific Reports*** (2021)
8. Murphy, S., ... **Kambhampati, S.**, *et al.* PGC1/PPAR Drive Cardiomyocyte Maturation through Regulation of Yap1 and SF3B2. ***Nature Communications*** (2021)

* or # denotes equal contribution

Oral Presentations (selected)

<i>Boston, USA</i>	Eric and Wendy Schmidt Center Symposium: Biomedical Science/AI	2025
<i>Seoul, South Korea</i>	Research in Computational Molecular Biology (RECOMB)	2025
<i>Vancouver, Canada</i>	Keystone Symposia: Single-cell Biology and Tissue Genomics	2024

Leadership and Outreach

Health Professions Recruitment Program – Mentor	2021 – 2023
Mentored underserved Boston -area high schoolers interested in science and medicine	
Hopkins Undergraduate Research Journal – Editor-in-Chief	2018 – 2021
- Published an annual journal of 15+ STEM and Humanities articles - Established a faculty review process and a mission on accessible scientific communication	
Thread – Head of Family	2017 - 2021
Managed a “family” of volunteers to support at-risk Baltimore high school students	

Teaching and Mentorship

Eric Wang (MIT undergraduate, Eric & Wendy Schmidt Center UROP)	2026
Qiyu Gong (Broad Institute, Associate Computational Biologist)	2022-2025
Ebtisam Alshehri (Broad Institute, Machine Learning Research Intern)	2024
Matthew Shum (Harvard undergraduate)	2022
Teaching Assistant: Systems and Controls (JHU BME Core Curriculum)	2019, 2020

Awards and Grants (selected)

<i>Anthropic AI for Science Grant</i> \$40k in Claude credits to develop a lab-in-the-loop automated drug discovery system for treating melanoma	2026
<i>NSF Graduate Research Fellowship Program (GRFP)</i> Covers PhD tuition+stipend for 3 years	2022
<i>Hertz Fellowship Finalist</i> 1 of 45 finalists selected from 650+ applicants	2022
<i>Tau Beta Pi Honor Society</i> Top 1/5 of JHU seniors in engineering	2021
<i>Frances Howard Flatau Scholarship</i> To 1 JHU BME upperclassman annually	2020, 2021

Hobbies and Interests

Rock climbing, Harvard Mountaineering Club, sailing, mountain biking, chess, Indian classical violin